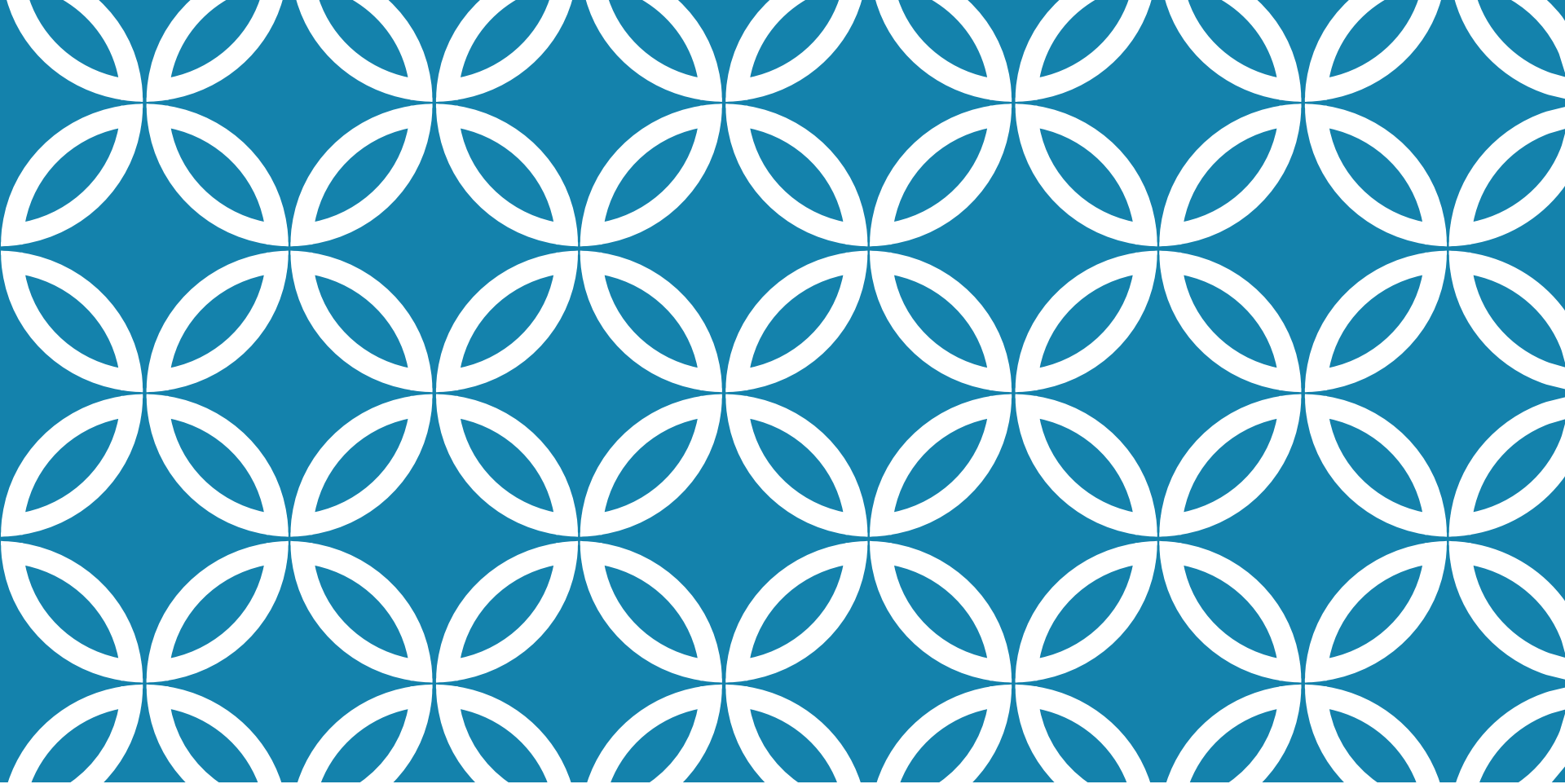




REAL ESTATE PRICE ANALYSIS

Exploratory Data Analysis
(EDA)

Sumeet Rajput

PROJECT OVERVIEW

- 1) Exploratory Data Analysis of real estate pricing data.
- 2) Focus on identifying key factors affecting house prices.
- 3) Dataset contains property features, prices, and amenities.

OBJECTIVES

- 1) Clean and preprocess real estate data
- 2) Analyze price distributions and relationships
- 3) Engineer new features for deeper insights
- 4) Understand market trends and customer preferences

TOOLS & TECHNOLOGIES USED

- 1) Python
- 2) Pandas & NumPy for data manipulation
- 3) Matplotlib & Seaborn for visualization
- 4) Google Colab for execution

DATA CLEANING

1. Removed duplicate records
2. Handled missing values using median imputation
3. Removed unnecessary columns
4. Ensured dataset consistency and quality

UNIVARIATE ANALYSIS

1. Analyzed distribution of house prices
2. Observed right-skewed price distribution
3. Identified presence of premium and luxury properties

MULTIVARIATE ANALYSIS

- 1.Used correlation matrix to analyze feature relationships
- 2.Strong correlation between price and living area
- 3.Overall quality and garage capacity significantly impact price

FEATURE ENGINEERING

- 1.Created Price per Square Foot feature
- 2.Calculated House Age at time of sale
- 3.Improved interpretability of pricing behavior

SIZE & FEATURE IMPACT

1. Larger living area leads to higher house prices
2. Number of bedrooms influences valuation
3. Size-related features are strong price drivers

MARKET TRENDS & HISTORICAL PRICING

1. Analyzed median sale price over years
2. Identified market fluctuations over time
3. Median price used to reduce impact of outliers

CUSTOMER PREFERENCES & AMENITIES

1. Garage capacity used as key amenity indicator
2. More amenities generally lead to higher prices
3. Customer preferences strongly affect valuation

KEY INSIGHTS

1. Living area and quality are dominant pricing factors
2. Amenities add significant value to properties
3. Feature engineering enhances analysis quality
4. Clear market trends observed over time

CONCLUSION

EDA revealed critical drivers of real estate pricing

Insights useful for buyers, sellers, and investors

Project provides strong foundation for predictive modeling



THANK YOU

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GitHub: [Real-Estate-EDA](#)

Questions & Discussion